

1. Design FSM with the given input alphabet that accepts the given set of strings:

a) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that start with 01.

b) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that end with 01.

c) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that contain the pattern 010.

d) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that start with 0 and contain exactly one 1.

e) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that start with 01 and end with 100.

f) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that do not contain two consecutive 0s.

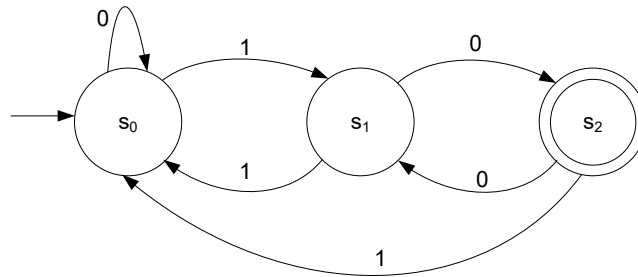
g) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that end with 100 or 01.

h) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that do not end with 100 and do not end with 01.

i) Input alphabet = $\{0, 1\}$; Accepts the set of all strings that contain 0101 but not 100.

j) Input alphabet = $\{0, 1\}$; Accepts the set of all strings containing an even number of 1s and an odd number of 0s.

2. Characterize the set of strings accepted by the following machine:



If all of the problems can't be done in the lab period, students can work on them on their own.